CHAPTER 9: ONE SAMPLE HYPOTHESIS TESTING			
	Lower tail test	**Two tail test**	**Upper tail test**
Hypotheses	**For the population mean:** $H_0: \mu = _$ $H_1: \mu < _$ **For the population proportion:** $H_0: \pi = _$ $H_1: \pi < _$	**For the population mean:** $H_0: \mu = _$ $H_1: \mu \neq _$ **For the population proportion:** $H_0: \pi = _$ $H_1: \pi \neq _$	**For the population mean:** $H_0: \mu = _$ $H_1: \mu > _$ **For the population proportion:** $H_0: \pi = _$ $H_1: \pi > _$
Test statistic	**For the mean, when σ is known:** $zstat = \frac{\bar{x} - \mu}{\sigma/n}$ **For the mean, when σ is unknown:** $tstat = \frac{\bar{x} - \mu}{\sigma/n}$ **For the proportion:** $zstat = \frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$	**For the mean, when σ is known:** $zstat = \frac{\bar{x} - \mu}{\sigma/n}$ **For the mean, when σ is unknown:** $tstat = \frac{\bar{x} - \mu}{\sigma/n}$ **For the proportion:** $zstat = \frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$	**For the mean, when σ is known:** $zstat = \frac{\bar{x} - \mu}{\sigma/n}$ **For the mean, when σ is unknown:** $tstat = \frac{\bar{x} - \mu}{\sigma/n}$ **For the proportion:** $zstat = \frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$
Critical Value	**z-test:** $\alpha \rightarrow$ [INV] [Z⇄P] 3 **t-test:** df → [INV] [df t⇄P] 2 → α → [=] where $df = n - 1$	**z-test:** $\frac{\alpha}{2} \rightarrow$ [INV] [Z⇄P] 3 **t-test:** df → [INV] [df t⇄P] 2 → $\frac{\alpha}{2}$ → [=] where $df = n - 1$	**z-test:** $1 - \alpha \rightarrow$ [INV] [Z⇄P] 3 **When t-test:** df → [INV] [df t⇄P] 2 → $1 - \alpha$ → [=] where $df = n - 1$
P-value	p-value $= P(z \text{ or } t < \text{test statistic})$ **z-test:** zstat → [Z⇄P] 3 **t-test:** df → [df t⇄P] 2 →tstat→ [=]	p-value $= 2 \cdot P(z \text{ or } t < -\text{test statistic})$ **z-test:** zstat → [Z⇄P] 3 **t-test:** df → [df t⇄P] 2 →tstat→ [=]	p-value $= P(z \text{ or } t > \text{test statistic})$ $= 1 - P(z \text{ or } t < \text{test statistic})$ **z-test:** zstat → [Z⇄P] 3 **t-test:** df → [df t⇄P] 2 →tstat→ [=]